

Randomized optimal efficiency predictive probability designs for early phase biomarker-targeted oncology trials

Emily C. Zabor

Department of Quantitative Health Sciences

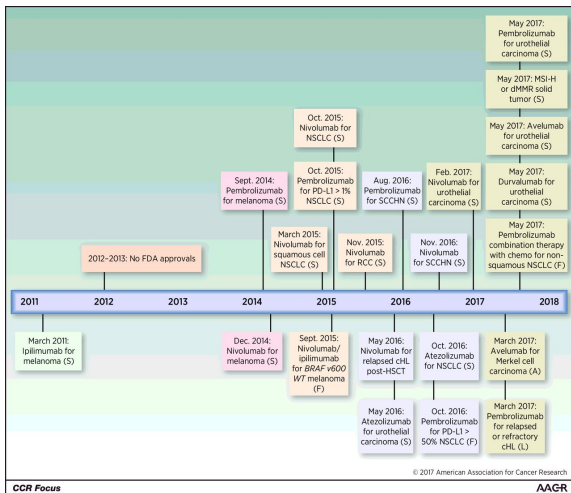
NESS - June 5, 2023

Early phase oncology trials are changing in response to the changing oncology treatment landscape

	Cytotoxic chemotherapy	Molecularly targeted agents	Immuno-stimulatory antibodies
Patients number	30-50 unselected patients Target enrichment	30-200 molecularly selected patients Immune enrichment	100-1000 immunologically selected patients Pts # Unselected Selected
Route of administration	IV > Oral	Oral > IV	Novel routes of administration (intra-tumoral)
Toxicity	MTD quasi-systematically reached	MTD unconsciously reached	MTD rarely reached → MAD
PK/PD - biomarkers	Traditional PK limited PD AUC OIB	Traditional PK with potential for PK - based dose recommendation Biomarker-driven PD for target assay validation and molecular enrichment OID?	PK and pD-based dose recommendation? repeated PD for dynamic biomarkers and immunological monitoring
Design	Traditional 3 + 3 dose-escalation design 20-30 pts Escalation : Expansion	3 + 3 dose-escalation design with large expansion cohorts in selected populations 30-300 selected pts Molecular enrichment Escalation : Expansion	Accelerated titration/adaptive design multiple parallel expansion cohorts long-term follow-up + drug rechallenge 100-1000 pts Escalation : Expansion +/- immune enrichment
Drug approval	Based on later phase 2 or 3 trials P1 P2 P3 Approval	Conditional of accelerated approval based on large molecularly selected expansion cohorts P1 P2 P3 Approval Conditional/accelerated approval	Conditional of accelerated approval based on histology and immune-biomarker selected expansion cohorts P1 P3 Approval Conditional/accelerated approval
Drug development timeframe	10 years	5-8 years	<5 years

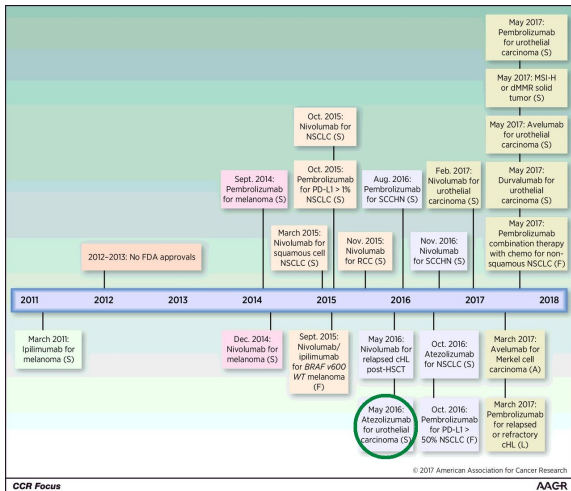
Postel-Vinay, S. et al. (2016). Annals of Oncology, 27(2), 214–224.

One area of rapid development is immune checkpoint inhibitors



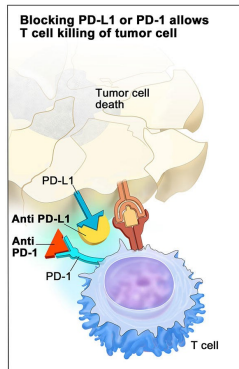
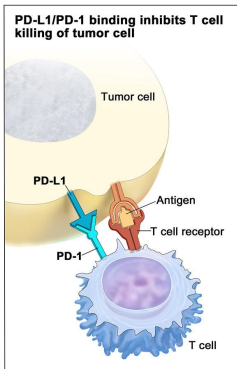
Baik, C. S. et al. (2017). Clinical Cancer Research, 23(17), 4992–5002.

Case study: Atezolizumab in urothelial carcinoma



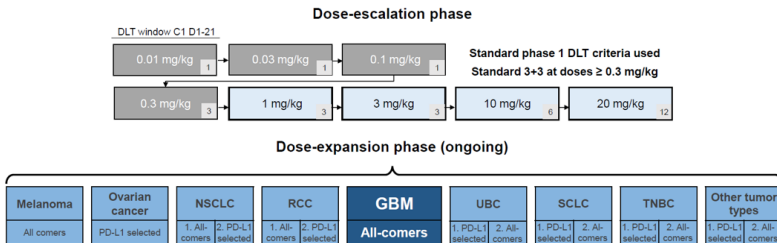
Baik, C. S. et al. (2017). Clinical Cancer Research, 23(17), 4992–5002.

Atezolizumab is a PD-L1 checkpoint inhibitor



<https://www.cancer.gov/news-events/cancer-currents-blog/2016/fda-atezolizumab-bladder>
<https://www.roche.com/solutions/pharma/productid-8c738504-101a-40b9-867b-d907c578df51>

Phase I study of atezolizumab included multiple locally advanced or metastatic solid tumor types



Supplement to: Lukas RV et al. (2018). Journal of Neuro-Oncology, 140(2):317-328.

Atezolizumab (MPDL3280A) Monotherapy for Patients With Metastatic Urothelial Cancer

Long-term Outcomes From a Phase 1 Study

Daniel P. Petrylak, MD; Thomas Powles, MBBS, MRCP, MD; Joaquim Bellmunt, MD, PhD; Fadi Braiteh, MD; Yohann Loriot, MD, PhD; Rafael Morales-Barrera, MD; Howard A. Burris, MD; Joseph W. Kim, MD; Beiying Ding, PhD; Constanze Kaiser, PhD; Marcella Fassò, PhD; Carol O'Hear, MD, PhD; Nicholas J. Vogelzang, MD

Table 3. Objective Response Rates to Atezolizumab Treatment and Duration of Response by Programmed Death Ligand 1 Immunohistochemical Status

Parameter	IC0/1 (n = 44)	IC2/3 (n = 50)	All Patients (N = 95) ^a
Objective response rate (95% CI) ^b	11 (4 to 25)	40 (26 to 55)	26 (18 to 36)
Best overall response, No. (%)			
Complete response	1 (2)	8 (16)	9 (10)
Partial response	4 (9)	12 (24)	16 (17)
Stable disease	9 (21)	9 (18)	18 (19)
Progressive disease	24 (55)	17 (34)	42 (44)
No assessment ^c	6 (14)	4 (8)	10 (11)
Duration of response, mo (range)	27.6 (6.2 to >34.3)	18.0 (2.8 to >41.0)	22.1 (2.8 to >41.0)

Petrylak, D. P. et al. (2018). JAMA oncology, 4(4), 537–544.

Atezolizumab in patients with locally advanced and metastatic urothelial carcinoma who have progressed following treatment with platinum-based chemotherapy: a single-arm, multicentre, phase 2 trial

Jonathan E Rosenberg, Jean Hoffman-Censits, Tom Powles, Michiel S van der Heijden, Arjun V Balar, Andrea Necchi, Nancy Dawson, Peter H O'Donnell, Ani Balmanoukian, Yohann Loriot, Sandy Srinivas, Margitta M Retz, Petros Grivas, Richard W Joseph, Matthew D Galsky, Mark T Fleming, Daniel P Petrylak, Jose Luis Perez-Gracia, Howard A Burris, Daniel Castellano, Christina Canil, Joaquim Bellmunt, Dean Bajorin, Dorothee Nickles, Richard Bourgon, Garrett M Frampton, Na Cui, Sanjeev Mariathasan, Oyewale Abidoye, Gregg D Fine, Robert Dreicer

	Patients, n	Objective response rate, n (%) [95% CI]	Complete response	Partial response	Stable disease	Progressive disease
RECIST version 1.1 criteria by independent review						
IC2/3	100	26 (26% [18–36])	11 (11%)	15 (15%)	16 (16%)	44 (44%)
IC1/2/3	207	37 (18% [13–24])	13 (6%)	24 (12%)	34 (16%)	107 (52%)
All patients	310	45 (15% [11–19])	15 (5%)	30 (10%)	59 (19%)	159 (51%)
IC1*	107	11 (10% [5–18])	2 (2%)	9 (8%)	18 (17%)	63 (59%)
IC0*	103	8 (8% [3–15])	2 (2%)	6 (6%)	25 (24%)	52 (50%)

Rosenberg, J. E. et al. (2016). Lancet (London, England), 387(10031), 1909–1920.



FDA Approval Summary: Atezolizumab for the Treatment of Patients with Progressive Advanced Urothelial Carcinoma after Platinum-Containing Chemotherapy

YANG-MIN NING, DANIEL SUZMAN, V. ELLEN MAHER, LIJUN ZHANG, SHENGHUI TANG, TIFFANY RICKS, TODD PALMBY, WENTAO FU, QI LIU, KIRSTEN B. GOLDBERG, GEOFFREY KIM, RICHARD PAZDUR

Center for Drug Evaluation and Research, U.S. Food and Drug Administration, Silver Springs, Maryland, USA

Key Words. Locally advanced or metastatic urothelial carcinoma • Bladder cancer • Platinum-containing chemotherapy •

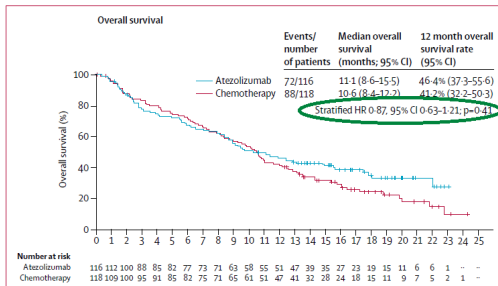
Atezolizumab • Immunotherapy

Early clinical studies showed that atezolizumab is active in numerous malignancies, including advanced urothelial carcinoma [9, 10]. Based on the preliminary clinical evidence showing a confirmed response rate of 50% (95% CI: 29.3%, 70.7%) in 20 patients with high PD-L1 expression as determined by the prototype assay, atezolizumab received a Breakthrough Therapy designation in May 2014 for the treatment of patients with advanced urothelial carcinoma who have received prior platinum-containing chemotherapy. The application also received priority review, and the proposed indication was approved in May 2016. Herein, we summarize key review findings that supported this approval.

Ning, Y. M. et al. (2017). The Oncologist, 22(6), 743–749.

Atezolizumab versus chemotherapy in patients with platinum-treated locally advanced or metastatic urothelial carcinoma (IMvigor211): a multicentre, open-label, phase 3 randomised controlled trial

Thomas Powles, Ignacio Durán, Michiel S van der Heijden, Yohann Loriot, Nicholas J Vogelzang, Ugo De Giorgi, Stéphane Oudard, Margitta M Retz, Daniel Castellano, Aristotelis Bamias, Aude Fléchon, Gwenaëlle Gravis, Syed Hussain, Toshimi Takano, Ning Leng, Edward E Kadel III, Romain Banchereau, Priti S Hegde, Sanjeev Mariathasan, Na Cui, Xiaodong Shen, Christina L Derleth, Marjorie C Green, Alain Ravaud



	IC2/3 population	
	Atezolizumab group (n=116)	Chemotherapy group (n=118)
Progression-free survival		
Patients with event (%)*	93 (80%)	105 (89%)
Median (months; 95% CI)	2.4 (2.1-4.2)	4.2 (3.7-5.0)
Objective response†		
Number of evaluable patients	113	116
Number of patients with response (%; 95% CI)	26 (23.0%, 15.6-31.9)	25 (21.6%, 14.5-30.2)

Powles, T. et al. (2018). Lancet (London, England), 391(10122), 748-757.

Atezolizumab's Indication in Previously Treated Metastatic Bladder Cancer Is Withdrawn

By The ASCO Post Staff

Posted: 3/8/2021 2:24:00 PM

Last Updated: 3/31/2021 3:32:00 PM

Atezolizumab Indication in US Withdrawn for Previously Treated Metastatic Urothelial Cancer

March 8, 2021

Audrey Sternberg

Genentech, a member of the Roche Group, is voluntarily withdrawing the U.S. indication for atezolizumab (Tecentriq) in patients with urothelial carcinoma following platinum-based chemotherapy. The U.S. Food and Drug Administration (FDA) has approved the withdrawal of the indication.

As part of an industry-wide assessment of indications based on accelerated approval, Roche in consultation with the FDA has decided to withdraw the indication for atezolizumab in patients with urothelial carcinoma following platinum-based chemotherapy.

Atezolizumab FDA indication for pretreated bladder cancer voluntarily withdrawn

March 8, 2021

Jason M. Broderick

The withdrawal comes after the confirmatory phase 3 IMvigor211 study missed its primary end point.

Roche (Genentech) has voluntarily withdrawn the FDA indication for the PD-L1 inhibitor atezolizumab (Tecentriq) for use in patients with locally advanced or metastatic urothelial carcinoma (mUC) previously treated with platinum-based chemotherapy.¹

<https://ascopost.com>; <https://www.cancernetwork.com>; <https://www.urologytimes.com>



Historical control rates are population averages and may not apply to biomarker-targeted groups

Phase II	Patients, n	Objective response rate, n (%) [95% CI]	Complete response	Partial response	Stable disease	Progressive disease
RECIST version 1.1 criteria by independent review						
IC2/3	100	26 (26% [18–36])	11 (11%)	13 (13%)	15 (15%)	2 (2%)
IC1/2/3	207	37 (18% [13–24])	2 (2%)	11 (10%)	15 (15%)	2 (2%)
All patients	310	45 (15% [11–19])	13 (13%)	24 (24%)	30 (30%)	5 (5%)
IC1*	107	11 (10% [5–18])	2 (2%)	11 (10%)	15 (15%)	2 (2%)
IC0*	103	8 (8% [3–15])	2 (2%)	11 (10%)	15 (15%)	2 (2%)

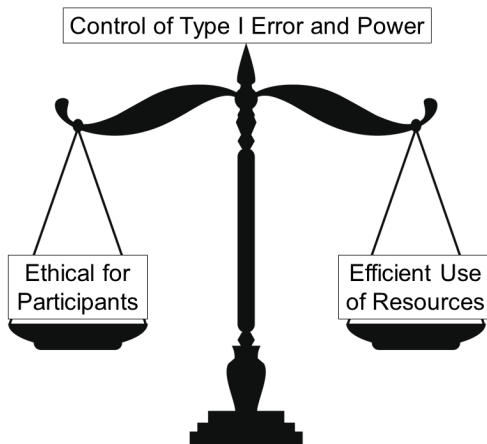
Phase III	IC2/3 population		ITT population	
	Atezolizumab group (n=116)	Chemotherapy group (n=118)	Atezolizumab group (n=467)	Chemotherapy group (n=464)
Progression-free survival				
Patients with event (%)*	93 (80%)	105 (89%)	407 (87%)	410 (88%)
Median (months; 95% CI)	2.4 (2.1–4.2)	4.2 (3.7–5.0)	2.1 (2.1–2.2)	4.0 (3.4–4.2)
Objective response†				
Number of evaluable patients	113	116	462	461
Number of patients with response (%; 95% CI)	26 (23.0%, 15.6–31.9)	25 (21.6%, 14.5–30.2)	62 (13.4%, 10.5–16.9)	62 (13.4%, 10.5–16.9)

Rosenberg, J. E. et al. (2016). Lancet (London, England), 387(10031), 1909–1920.
 Powles, T. et al. (2018). Lancet (London, England), 391(10122), 748–757.

With the **large sample sizes** employed across phase I expansion and phase II - **405 patients total** - could randomization have been employed earlier?

Goal: Develop a **randomized** design for early phase trials of **biomarker-targeted subgroups** using the **optimal efficiency** predictive probability method.

Futility stopping can be used as a mechanism to achieve multiple expansion cohort goals



Consider the setting of a two-arm clinical trial with a binary response endpoint

Each patient i either has the outcome (or not) such that $x_i = 1(0)$. Then $X = \sum_{i=1}^n x_i$ is the total number of responses out of n patients observed so far of N patients total.

We test:

$$H_0 : p_1 \leq p_0$$

vs

$$H_1 : p_1 > p_0$$

Bayesian statistical paradigm is the basis for sequential statistical learning

Beta-binomial model:

Prior: $\pi(p) \sim \text{Beta}(a_0, b_0)$ [Specifically $\text{Beta}(0.5, 0.5)$ here]

Likelihood: $L_x(p) \propto p^x (1 - p)^{n-x}$

Posterior: $p|x \sim \text{Beta}(a_0 + x, b_0 + n - x)$

The treatment is **efficacious** at **posterior threshold θ** if:

$$\Pr(p_1 > p_0 | X) > \theta$$

Predictive probability monitoring stops the trial according to a threshold

The posterior predictive distribution of the number of future responses X^* in the remaining $n^* = N - n$ future patients follows:

$$\text{Beta} - \text{binomial}(n^*, a_0 + x, b_0 + n - x)$$

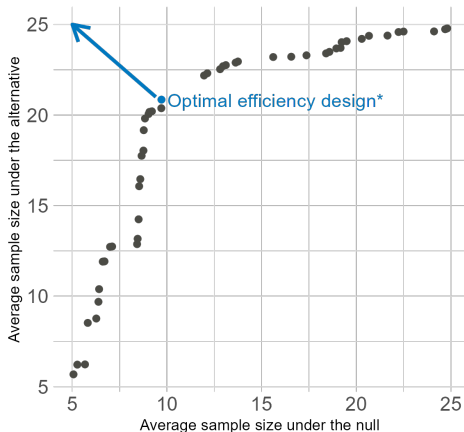
$$PPP = \sum_{x^*=0}^{n^*} Pr(X^* = x^* | x) \times I(Pr(p_1 > p_0 | X, X^* = x^*) > \theta)$$

We stop for **futility** at **predictive threshold θ^*** if **$PPP < \theta^*$** .

Simulation-based methods evaluate a grid of thresholds to compare operating characteristics

	θ^*			
θ	0.05	0.10	0.15	0.20
0.90				
0.91				
0.92				
0.93				
0.94				
0.95				
0.96				
0.97				
0.98				
0.99				

The optimal efficiency criteria is used to calibrate posterior and predictive thresholds



*Subject to constraints on type I error and power

Zabor, E. C. et al. (2022). JCO Precision Oncology, 6, e2100390.

The optimal efficiency design minimizes the defined Euclidean distance, with optional weighting

Define efficiency Euclidean distance (EED):

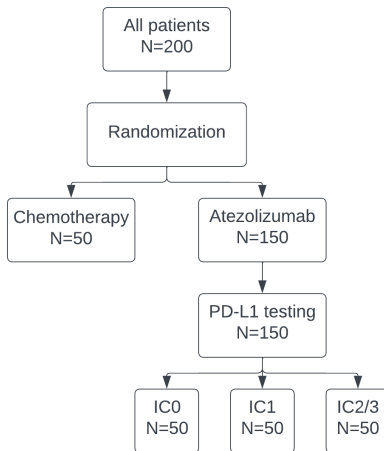
$$EED_{km} = w_{\bar{N}_{H_0}} * (\bar{N}_{H_0 km} - \min(\bar{N}_{H_0}))^2 + w_{\bar{N}_{H_1}} * (\bar{N}_{H_1 km} - \max(\bar{N}_{H_1}))^2$$

Zabor, E. C. et al. (2022). JCO Precision Oncology, 6, e2100390.

Three randomized design options are based on the atezolizumab study

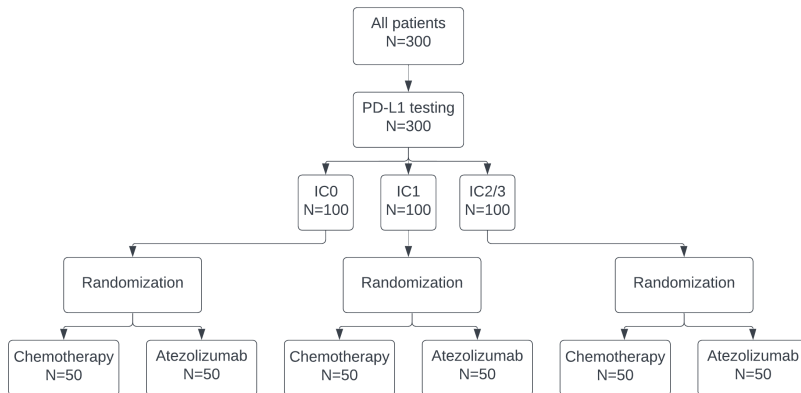
- Three biomarker subgroups have equal population prevalence of 33%
- Null response rate based on historical control rate of 10%
- Subgroup-specific alternative response rates of 10%, 20%, and 30% in the IC0, IC1, and IC2/3 subgroups
 - ▶ i.e. biomarker is hypothesized to be predictive
- Presented results conduct **interim looks for futility** after every **10 patients**

Pooled control arm design



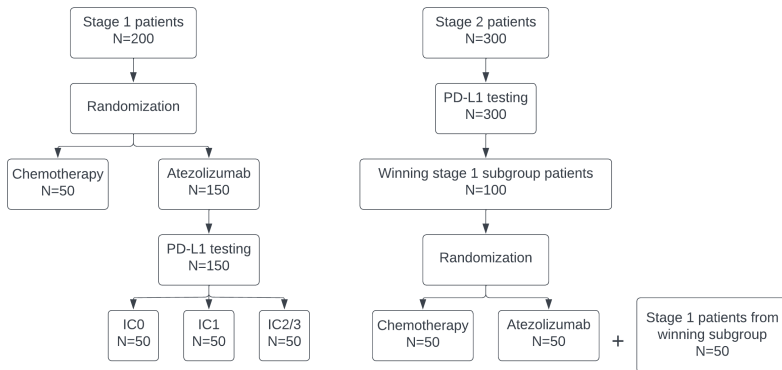
Zabor EC, Kaizer AM, Pennell NA, Hobbs BP. *Frontiers in Oncology*. 2022;12:955056.

Stratified control arm design



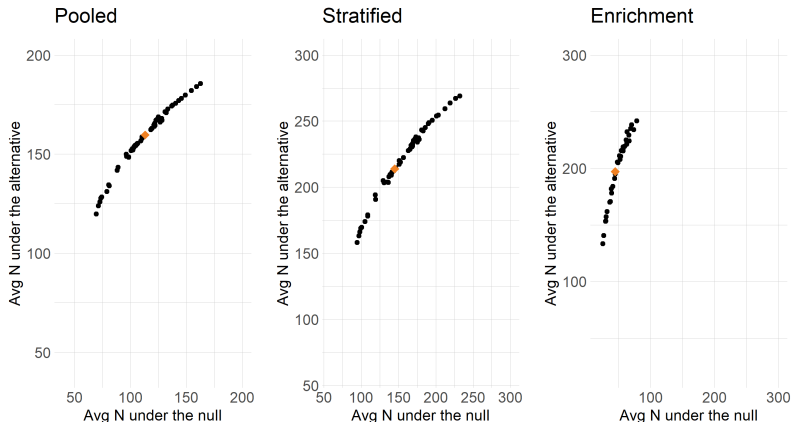
Zabor EC, Kaizer AM, Pennell NA, Hobbs BP. *Frontiers in Oncology*. 2022;12:955056.

Enrichment design



Zabor EC, Kaizer AM, Pennell NA, Hobbs BP. *Frontiers in Oncology*. 2022;12:955056.

Optimal efficiency designs were selected from design options with type I error 0.05-0.1 and power ≥ 0.8



Zabor EC, Kaizer AM, Pennell NA, Hobbs BP. *Frontiers in Oncology*. 2022;12:955056.

All three designs lead to reasonable type I error and power to detect the subgroup of interest

	Pooled		Stratified		Enrichment			
	Type I	Power	Type I	Power	Stage 1		Stage 2	
					Type I	Power	Type I	Power
IC0	0.05	0.05	0.05	0.08	—	—	—	—
IC1	0.05	0.40	0.08	0.45	—	—	—	—
IC2/3	0.07	0.80	0.07	0.82	—	—	—	—
Overall	—	—	—	—	0.09	0.73	0.09	0.86

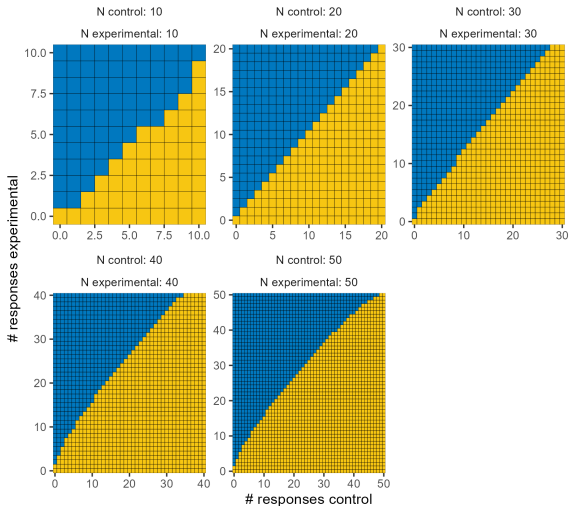
Zabor EC, Kaizer AM, Pennell NA, Hobbs BP. *Frontiers in Oncology*. 2022;12:955056.

All designs result in reasonable sample sizes compared to those actually used in phases I and II

	Pooled		Stratified		Enrichment	
	N Null	N Alt	N Null	N Alt	N Null	N Alt
Control	35	48	73	107	33	81
IC0	26	27	24	24	—	—
IC1	27	39	25	38	—	—
IC2/3	27	47	25	47	—	—
Atezo	79	112	73	107	68	137
Total	114	160	145	214	101	218

Zabor EC, Kaizer AM, Pennell NA, Hobbs BP. *Frontiers in Oncology*. 2022;12:955056.

Seamless implementation with the creation of design-stage decision tables



Decision ■ Proceed ■ Stop

Selection of the ideal design depends on the clinical setting

	Pooled	Stratified	Enrichment
Smallest sample size	✓		
Predictive biomarker	✓	✓	✓
Prognostic biomarker		(?)	
Most biomarker testing			✓
Least biomarker testing	✓		

The stratified control arm design averages 214 patients under the alternative, and only 145 patients under the null, as compared to the 95 used in the phase I expansion followed by 310 used in phase II.

In the context of this case study, using this design would also avoid enrolling the 931 patients who were studied in phase III based on incomplete early phase results.

R package {ppseq} implements optimal sequential predictive probability designs

ppseq 0.2.0  Reference Articles ▾ News

ppseq

The {ppseq} package provides functions to design clinical trials using Bayesian sequential predictive probability monitoring. Functionality is available to design [one-arm](#) or [two-arm](#) trials by searching over a grid of combinations of posterior and predictive thresholds and identifying the optimal design according to two criteria: accuracy and efficiency. Interactive plotting allows easy comparison of the various design options and easy trial implementation through decision rule plots.

Installation

You can install the production version of ppseq from CRAN with:

```
install.packages("ppseq")
```

Or you can install the development version of ppseq from GitHub with:

```
remotes::install_github("zabore/ppseq")
```

Links

[View on CRAN](#)
[Browse source code](#)
[Report a bug](#)

License

[MIT](#) + file [LICENSE](#)

Citation

[Citing ppseq](#)

Developers

Emily C. Zabor
Author, maintainer 

Brian P. Hobbs
Author

Michael J. Kane
Author 

Zabor, EC, Hobbs, BP, Kane, MJ. (2023). The R Journal, 14(4), 280-290.
<http://www.emilyzabor.com/ppseq/>

Thank you! Questions?

✉ zabore2@ccf.org

🐦 [@zabormetrics](https://twitter.com/zabormetrics)

🗣️ [@zabore](https://www.youtube.com/channel/UCzabore)

🌐 www.emilyzabor.com